



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name: Janushiya.P.A

Reg No: 1924260368

Course Code: DSA0401

Course Name: FUNDAMENTAL OF DATASCIENCE

3. Calculate the average sale price of houses with more than four bedrooms

AIM:

To find the average sale price of houses having more than four bedrooms using NumPy filtering.

ALGORITHM:

1. Import the NumPy library.
2. Create an array containing bedrooms, square footage, and sale price.
3. Select houses with more than four bedrooms.
4. Calculate the average of their sale prices.
5. Display the result.

CODE:

```
import numpy as np
```

```
house_data = np.array([
```

```
[3, 1500, 250000],  
[5, 2200, 400000],  
[6, 2800, 500000],  
[4, 1800, 300000]  
])
```

```
selected = house_data[house_data[:, 0] > 4]
```

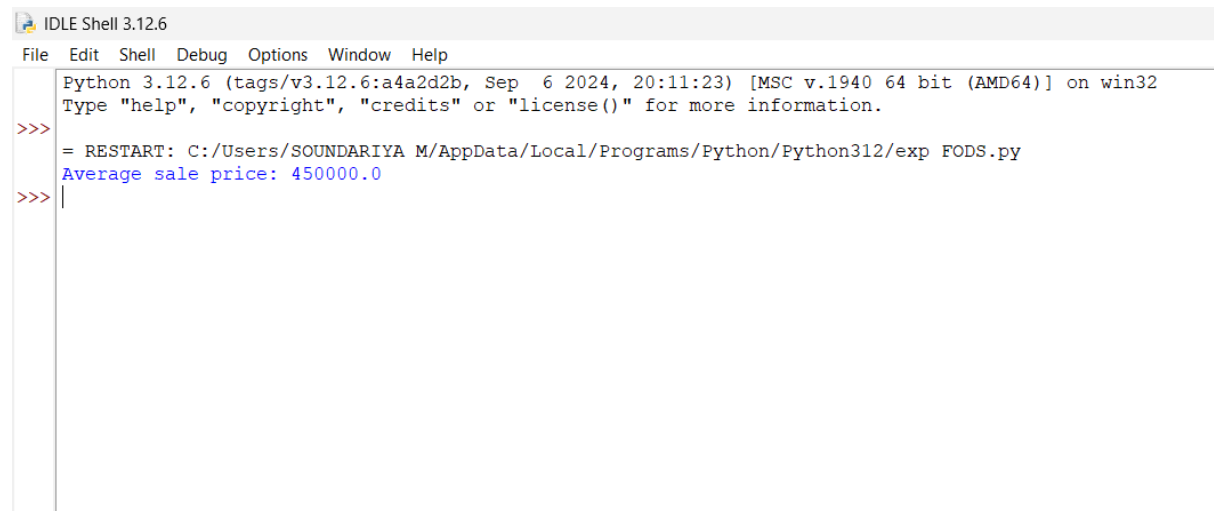
```
average = np.mean(selected[:, 2])
```

```
print("Average sale price:", average)
```

INPUT:

Each row contains [bedrooms, square_feet, sale_price].

OUTPUT:



```
IDLE Shell 3.12.6  
File Edit Shell Debug Options Window Help  
Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep 6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32  
Type "help", "copyright", "credits" or "license()" for more information.  
>>> = RESTART: C:/Users/SOUNDARIYA M/AppData/Local/Programs/Python/Python312/exp FODS.py  
Average sale price: 450000.0  
>>> |
```

RESULT:

The average sale price of houses with more than four bedrooms is 450,000.